

Functional Requirements Document

Status: Product development | Domain: Event technology and payments | Role: Founder and Product Business Analyst

Purpose

Describes how the solution must behave so each business requirement can be designed, built and tested. Each functional requirement traces to a parent business requirement.

Functional requirements

ID	Parent	System behaviour	Business rule or validation
FR-01	BR-01	Event wizard captures type, dates, capacity and ticket tiers	Recurring events require an end date or occurrence count
FR-02	BR-02	Ticket record holds owner, state and state history	States: Reserved, Issued, Transferred, Refunded, Scanned, Void
FR-03	BR-03	Transfer requires recipient acceptance before ownership moves	Scanned or refunded tickets cannot be transferred
FR-04	BR-03	Refund eligibility checks organiser policy and event date	Refund voids the ticket before money is returned
FR-05	BR-04	Scanner validates ticket state in real time	Valid scan moves ticket to Scanned; repeat scan shows first-scan time
FR-06	BR-05	Settlement report totals sales, fees, refunds and chargebacks per event	Payout blocked while variance is not zero or explained

User stories

ID	Parent	Story	Acceptance (Given / When / Then)
US-01	BR-03	As a attendee, I want to transfer my ticket to a friend, so that they can attend in my place.	Given I own an Issued ticket When my friend accepts the transfer Then the ticket shows my friend as owner and my copy is void
US-02	BR-04	As a door staff member, I want a clear result when I scan, so that I admit only valid tickets quickly.	Given a ticket was already scanned When I scan it again Then the scanner shows Rejected with the first-scan time
US-03	BR-05	As a organiser, I want to see reconciled totals before payout, so that I trust what I am paid.	Given my event has ended When I open settlement Then sales, fees, refunds and variance are shown before I confirm

Non-functional requirements

Area	Requirement
Performance	Scan result returned in under two seconds
Security	Ticket codes cannot be guessed or reused
Auditability	Every state change stores who, when and why
Resilience	Offline scan mode syncs when connection returns

Key data

Field	Type	Allowed values
event_id	Text	EVT-0001 format
ticket_id	Text	Random, non-sequential
owner_id	Text	User ID
ticket_state	Category	Reserved, Issued, Transferred, Refunded, Scanned, Void

Field	Type	Allowed values
payment_status	Category	Pending, Confirmed, Failed, Refunded
scan_timestamp	Datetime	ISO 8601